PEARSON EDEXCEL INTERNATIONAL GCSE · FURTHER PURE MATHEMATICS · 4PM1/02

A* Examination Guide

Built from 16 past papers and their mark schemes, January 2019 to June 2026

This guide is built from the 4PM1 Paper 2 question papers and mark schemes themselves. Methods are set out the way the mark schemes award M and A marks, and every warning comes from a note that examiners applied: “A0”, “withhold”, “do not allow”, “answers with no working score no marks”. Use the formula booklet for what it gives you, and learn the rest.

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Papers analysed: Jan 2019 (legacy 4PM0), Jun 2019, Jan 2020, Nov 2020, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 24).

1 HOW PAPER 2 IS BUILT

Format: 2 hours, 100 marks, 10 or 11 questions. The specification is the same as Paper 1, so any topic can appear, but Paper 2 leans harder on **trigonometric proof**, **rational curves with tangents and normals**, “**show that d^2y/dx^2 …**” questions, and **set-of-values inequalities**. The front page says “You must write down all the stages in your working”; the same formula sheet is printed on page 2.

Topic	Papers (of 16)	What is asked most
Integration: volumes and areas	14	volume about x- or y-axis, region between curve and line, exact form
Series (AP, GP, Σ)	13	find r and a, convergence, least n using logs, Σ notation
Trig identities and equations	11	derive double/triple angles from page 2, prove, then solve in a range
Vectors	10	parallelograms and trapezia, ratio of a line, area ratios, unit vectors
Rational curves, tangents, normals	9	asymptotes, intercepts, sketch, tangent parallel to a line, normal
Roots of quadratics (α, β)	8	symmetric functions, new equations, find p from a condition
Kinematics	8	v = ds/dt, a = dv/dt, at rest, distance, return to origin
Sectors, radians, segments	8	arc, sector, segment, overlapping circles, composite logos
Rates of change	7	spheres, cones, cubes, prisms; exact rates
3D trigonometry	7	pyramids, cubes, prisms; line–plane and plane–plane angles
Graphical solution of equations	7	table, curve, then the straight line to draw
Inequalities and the discriminant	8	linear and quadratic sets, “both”, real/unequal roots

Topic	Papers (of 16)	What is asked most
Optimisation	6	cylinder, prism, basket, shape of perimeter 18: max/min with justification
Logarithms	6	mixed bases, substitution u = log x, simultaneous logs
Binomial expansion	5	$(a + bx)^n$, product with another factor, integrate to estimate
"Show that" second-derivative relations	5	$y = e^{ax}$ sin bx or cos bx, find M and N

The recurring long questions

- **Trig chain (9–16 marks):** derive an identity from page 2 → use it to prove a result → solve an equation → integrate exactly (Jan 19, Nov 20, Jun 25, Jun 26).
- **Rational curve (10–14):** asymptotes and intercepts → sketch → gradient → tangent parallel to a line or normal through a point → a length or area (Jan 22, Nov 24, Jun 25, Jun 26).
- **Vectors (11–17):** prove a shape is a parallelogram or trapezium → collinear points → ratio → area ratio.
- **Volume or area (7–12):** find intersections, then integrate algebraically and give an exact answer.

2 TWENTY MARK-SCHEME RULES

1. **Answers only score nothing** where the question says "use algebra", "use calculus" or "using an algebraic method": "Answers only (potentially from a graphical calculator) scores B0 M0 A0 A0" (Jun 26).
2. **Calculator roots of a quadratic:** credit only if the quadratic is correct; "Incorrect quadratic solved by calculator M0A0" (Nov 20).
3. **"Using a formula from page 2, show that …":** start from the printed formula (e.g. cos(A + B) with B = A) and show each step.
4. **Exact means exact:** "Do not allow a mixture of decimals and exact values" (Jan 23); exact π answers as multiples of π (Jun 25).
5. **Radians when asked:** if the final answer is in degrees, both A marks are lost (Jan 19, Nov 20).
6. **Extra solutions in range score A0;** a value outside the range must not be rounded into it (Jun 23).
7. **Reject impossible values with a reason:** p = −5, k = −2 or a second t left in without rejection loses the final mark (Jun 22, Jan 23, Nov 20).
8. **$|r| < 1$ before S_∞:** using the sum to infinity with $|r| > 1$ loses the mark (Jan 23).
9. **Least n with logarithms:** "Candidates must use logarithms and we must see all previous method marks" (Nov 25).
10. **Algebraic integration must be seen** ("otherwise no marks", Jan 20); show the substitution of limits if the final answer is wrong.
11. **Correct limits:** "Do not accept limits of x = −2 and 4 or 6" where the region needs others (Nov 23).
12. **Second-derivative "show that":** substitute y, dy/dx and d^2y/dx^2 and simplify fully to the given result.
13. **Inequalities:** strict vs non-strict matters ("do not allow < for ≤", Nov 25); give the combined set clearly.
14. **Graph roots need the line drawn;** "without working these roots score no marks" (Nov 21).
15. **Asymptote sketches:** branches must not cross or bend back from the asymptotes; label intercepts.
16. **Factorising must be seen** when the question says "use algebra" (e.g. (3x − 4)(x + 5), Nov 25).
17. **Brackets:** "Do not allow missing brackets unless recovered later – this is a general point of marking."
18. **Given results exactly as written** (e.g. $S = 40x^2 + 1350/x$, Jun 22).
19. **Vector proofs:** show the parallel sides as scalar multiples and say which sides; "dividing by constants and then stating AD = BC is not acceptable" (Jun 26).

20. **Accuracy:** 3 s.f. by default; to the stated d.p. or s.f. otherwise (e.g. 8 s.f. in a binomial integral, Jun 26).

3 INEQUALITIES, QUADRATICS AND ROOTS

SET OF VALUES (JAN 20, JUN 21, JAN 23, JUN 25, NOV 25)

Linear: collect x terms; reverse the inequality when dividing by a negative. Quadratic: find the critical values, sketch the parabola, choose inside (< 0) or outside (> 0). "Both": overlap on a number line.

Discriminant (Jun 22): $2kx^2 + 5kx + 5k - 3 = 0$ has real roots $\Rightarrow 25k^2 - 8k(5k - 3) \geq 0 \Rightarrow 3k(8 - 5k) \geq 0 \Rightarrow 0 < k \leq 8/5$ ($k \neq 0$).

ROOTS A AND B

$\alpha + \beta = -b/a$, $\alpha\beta = c/a$; $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$; $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$; $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$; $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$.

Jun 22: $2x^2 + (6 + 2p)x + 2p = 0$: $\alpha + \beta = -(3 + p)$, $\alpha\beta = p$, $(\alpha - \beta)^2 = 9 + 2p + p^2$. If $\alpha - \beta = 3$, $p^2 + 2p = 0 \Rightarrow p = 0$ or -2.

New equation: $x^2 - (\text{sum})x + (\text{product}) = 0$, then clear fractions to integer coefficients.

- Completing the square (Jun 21, Jan 23): $A - B(x + C)^2$ gives a maximum A at $x = -C$.
- Factor theorem (Jun 21, Jun 24, Nov 24, Nov 25): $f(a) = 0$; one intersection with the x-axis is shown by a quadratic factor with negative discriminant (Jun 24).
- Real x for all y (Nov 20): rearrange to a quadratic in x and require its discriminant ≥ 0.

4 LOGARITHMS AND SURDS

WORKED EXAMPLES

Jun 22: $\log_4 x + \log_{16} x + \log_2 x = 10.5 \Rightarrow (\frac{1}{2} + \frac{1}{4} + 1)\log_2 x = 10.5 \Rightarrow \log_2 x = 6 \Rightarrow x = 64$.

Jan 19: $3\log_3 x - 8\log_x 3 = 10$: with $u = \log_3 x$, $3u - 8/u = 10 \Rightarrow (3u + 2)(u - 4) = 0 \Rightarrow x = 81$ or $3^{-2/3}$.

Nov 20: $16\log_r 4 = \log_4 r \Rightarrow (\log_4 r)^2 = 16 \Rightarrow r = 256$ or $1/256$.

- $\log_b a = 1/\log_a b$, and $\log_{a^n} x = (1/n)\log_a x$: convert everything to one base first.
- Simultaneous log equations (Jun 21, Nov 25): turn each into a power or product equation, then substitute.
- Surds (Jun 23, Jun 25, Jun 26): rationalise with the conjugate and show every line; the answer form $a\sqrt{b} - c\sqrt{d}$ tells you what to aim for.

5 SERIES

WORKED EXAMPLES

Jun 25: $ar^2 = 10$, $ar^6 = 40 \Rightarrow r^4 = 4 \Rightarrow r = \pm\sqrt{2}$. Second term negative $\Rightarrow r = -\sqrt{2}$, $a = 5$. $|r| > 1$, so there is no sum to infinity.

Jan 20: $t_3 = 24$, $t_2 + t_3 + t_4 = -36 \Rightarrow 2r^2 + 5r + 2 = 0 \Rightarrow r = -\frac{1}{2}$ (convergent) $\Rightarrow a = 96$, $S_\infty = 64$.

Least n (Nov 24, Nov 25): write $S_\infty - S_n$ or S_n as an inequality in r^n, take logs, reverse the sign when dividing by $\log r < 0$, and round up.

- Σ notation (Nov 20, Jun 26): $\sum_{r=1}^{n}(3r + 5) = n(3n + 13)/2$; for a sum from r = 35 subtract the sum to 34.
- Three terms in GP (Nov 20, Jan 23): $(\text{middle})^2 = \text{first} \times \text{third}$ gives an equation in the unknown.

6 BINOMIAL EXPANSION

METHOD

$(a + bx)^n = a^n(1 + bx/a)^n$, valid for $|x| < |a/b|$. Coefficients as exact fractions in lowest terms.

Nov 21: $3/(9 - 3x) = \frac{1}{3}(1 - x/3)^{-1} = \frac{1}{3} + x/9 + x^2/27 + x^3/81 + \ldots$, valid for $|x| < 3$.

Then multiply by another factor, e.g. $(1 + 2x)$ (Nov 21), and integrate term by term for an estimate to the stated accuracy (6 s.f. in Nov 21, 8 s.f. in Jun 26).

- Replacing x by x^2 or $-x$ in a known expansion (Jun 24, Jun 26): write it down directly, no new working.
- Unknown constants (Nov 20, Jun 24): set up the coefficient condition ($c_4 = 2c_3$) and solve.

7 GRAPHS AND COORDINATE GEOMETRY

GRAPHICAL SOLUTION (JAN 19, JUN 21, NOV 21, JAN 22, JUN 24, NOV 24, JUN 25)

Make the given equation look like "plotted curve = straight line". **Nov 24:** with $y = \log_{10}(6x - 1) - x$ plotted, the equation rearranges to $y = kx - 2$, so draw that line and read the intersection (2.0 or 2.1; "2" is not accepted).

Use the graph to estimate a value (ln 10.6 from $y = \ln(3x + 1) + 2$, Jan 19): show the lines you read from.

- Linear programming (Jun 23, Nov 23, Jun 26): draw, shade R, then test the objective at the vertices; with "integer coordinates" test lattice points near the optimal vertex.
- Coordinate geometry (Jun 21, Nov 21, Jun 22, Jun 23, Jun 26): ratio points, perpendicular lines, lengths, and areas of quadrilaterals; a right angle in a semicircle (Jun 23).

8 RATIONAL CURVES, TANGENTS AND NORMALS

TOOLKIT FOR $Y = (AX + B)/(CX + D)$

Asymptotes $x = -d/c$ and $y = a/c$ (as equations). Intercepts $(-b/a, 0)$ and $(0, b/d)$. Gradient $dy/dx = (ad - bc)/(cx + d)^2$ (quotient rule, on the sheet).

Tangent parallel to a line of gradient m: solve $(ad - bc)/(cx + d)^2 = m$, giving two points.

Normal through a point: gradient $-1/(dy/dx)$. Unknown constants: use the point and the gradient together.

WORKED EXAMPLE · JAN 22 Q10

$y = (2x - 1)/(x + 4)$: asymptotes $y = 2$, $x = -4$; intercepts $(½, 0)$, $(0, -¼)$; $dy/dx = 9/(x + 4)^2$.

Tangents parallel to $y = x$: $(x + 4)^2 = 9 \Rightarrow x = -1$ or -7. $P(-1, -1) \Rightarrow k_1 = 0$; $Q(-7, 5) \Rightarrow k_2 = 12$.

- Stationary points of a quadratic-over-linear curve (Nov 20): quotient rule, numerator = 0, then sketch with the vertical asymptote.
- Exponential curves (Nov 20, Nov 24, Jan 23): tangents meeting the axes; exact coordinates with ln.
- **No stationary points (Jun 26):** $f'(x) = 2e^{2x}(x^2 + x + 1)$; $e^{2x} > 0$ and $x^2 + x + 1$ has discriminant $-3 < 0$, so $f'(x) \neq 0$.

9 TRIGONOMETRY: IDENTITIES AND EQUATIONS

DERIVE FROM PAGE 2 (ASKED ON MOST PAPERS)

$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$; $\sin 2A = 2 \sin A \cos A$; $\tan 2A = 2 \tan A/(1 - \tan^2 A)$.

$\cos 3A = 4\cos^3 A - 3\cos A$ (from $\cos(2A + A)$, Jun 26); $\cos P + \cos Q = 2 \cos((P + Q)/2) \cos((P - Q)/2)$ (Nov 20); $\cos^4\theta - \sin^4\theta = \cos 2\theta$ (Jan 20, Jun 25).

WORKED EXAMPLES · EQUATIONS

Jun 25: $3\tan A = 2\sin A \Rightarrow \sin A(3 - 2\cos A) = 0 \Rightarrow \sin A = 0$ ($\cos A = 3/2$ impossible) $\Rightarrow A = 0°, 180°, 360°$. Dividing by $\sin A$ loses solutions.

Jun 21: $7\sin^2\theta + \sin\theta\cos\theta = 6 \Rightarrow \div\cos^2\theta$: $\tan^2\theta + \tan\theta - 6 = 0 \Rightarrow \tan\theta = 2$ or $-3 \Rightarrow 63°, 108°, 243°, 288°$.

Jun 26: $9\cos A - 12\cos^3 A - 2 = 0 \Rightarrow \cos 3A = -2/3 \Rightarrow$ for $-\pi \le A \le 0$, $A = -0.8, -1.3, -2.9$.

Jan 22: $\sin(2\theta - 20)° - \sqrt{3}\cos(2\theta - 20)° = 0 \Rightarrow \tan(2\theta - 20)° = \sqrt{3} \Rightarrow \theta = 40°, 130°$.

- Widen the range for multiple angles first; list all solutions, then remove any outside the range.
- Exact values (Jan 20): $\cos 15° = \cos(45° - 30°) = (\sqrt{6} + \sqrt{2})/4$; use 30°, 45°, 60° results, no calculator.
- Integrals of trig (Nov 20, Jun 25, Jun 26): use the identities to turn products and powers into sums first; $\cos^3\theta = (\cos 3\theta + 3\cos\theta)/4$.

10 RADIANS, TRIANGLES AND 3D

SECTORS

Arc $r\theta$, sector $\frac{1}{2}r^2\theta$, segment $\frac{1}{2}r^2(\theta - \sin\theta)$, θ in radians.

Nov 25: $r = 14$, perimeter 49 $\Rightarrow$ arc 21 $\Rightarrow$ area $\frac{1}{2} \times 14 \times 21 = 147\ \text{cm}^2$. **Nov 24:** $r = 2$, area $\pi/2 \Rightarrow \theta = \pi/4$, $P = 4 + \pi/2$.

Overlapping circles (Jun 22): $R = 2$ segments with $\theta = 2\pi/3 \Rightarrow P = 24\pi - 18\sqrt{3}$.

TRIANGLES AND 3D

Sine and cosine rules; area $\frac{1}{2}ab\sin C$. A cosine-rule quadratic (Jan 23, Jun 25) gives exact surd answers.

Angle between a line and a plane: use the line's projection. Angle between planes: drop perpendiculars to the common edge from each plane and use the triangle formed; obtuse angle when asked (Jan 22, Nov 25).

11 VECTORS

WORKED EXAMPLE · JAN 22 Q9

OA = 2a, OB = 4b, P the midpoint of AB: OP = a + 2b. OQ = ¾OP, so AQ = −(5/4)a + (3/2)b.

R on OB with A, Q, R collinear: OR = 2a + λ(−(5/4)a + (3/2)b); the a-coefficient is 0 ⇒ λ = 8/5 ⇒ OR = (12/5)b, so OR : RB = 3 : 2.

- Parallelogram (Jan 20, Jun 24): show AB = DC (or AD = BC) as equal vectors. Trapezium (Nov 25, Jun 26): one pair of sides parallel (scalar multiples) and the other pair not.
- $|\mathbf{a}| = |\mathbf{b}| = 1$ and $\mathbf{a} \perp \mathbf{b}$ (Nov 24, Nov 25): $|p\mathbf{a} + q\mathbf{b}| = \sqrt{(p^2 + q^2)}$; use this for unit vectors and lengths.
- Area ratios from vector ratios: triangles with a common height have areas in the ratio of their bases.

12 CALCULUS: RATES, OPTIMISATION, KINEMATICS

SECOND-DERIVATIVE "SHOW THAT" (JAN 20, JAN 23, JUN 24, JUN 25)

$y = e^{2x}\cos 2x$: $dy/dx = 2y - 2e^{2x}\sin 2x$; differentiate again and substitute to show $d^2y/dx^2 - 4dy/dx + 8y = 0$.

$y = e^{3x}\sin 2x$ satisfies $d^2y/dx^2 - 6dy/dx + 13y = 0$. Express everything in y, dy/dx and the leftover term.

RATES OF CHANGE

Nov 24: $V = 6h^3$, $dV/dt = -36$, $V = 384 \Rightarrow h = 4$; $dh/dt = -36 \div 18h^2 = -1/8$ cm/s (exact).

Jan 20: sphere, $dV/dt = 40$, $r = 4 \Rightarrow dr/dt = 5/(8\pi) \Rightarrow dS/dt = 8\pi r \times dr/dt = 20$ cm^2/s.

Small changes (Nov 21): $\delta y/y \times 100$ with $\delta x = 0.03x$ for a 3% change.

OPTIMISATION

Jun 25: total surface area $700\pi \Rightarrow V = \pi r(350 - r^2)$; $dV/dr = \pi(350 - 3r^2) = 0 \Rightarrow r = 10.8$; $d^2V/dr^2 = -6\pi r < 0$, so maximum; $h = 21.6$. **Nov 25** (322π): $r = 7.33$, $V_{max} = 2470$ cm^3.

Always justify with the sign of the second derivative and substitute back for the value asked.

KINEMATICS

$v = ds/dt$, $a = dv/dt$; $s = \int v\,dt + c$. **Nov 20:** $s = 3 + 8t + t^2 - t^3/3 \Rightarrow v = 8 + 2t - t^2 = 0 \Rightarrow t = 4 \Rightarrow s = 89/3$ m. Reject the negative time with a reason.

Minimum speed (Nov 25): complete the square on v or set $dv/dt = 0$. Return to origin (Nov 23): $s = 0$, $t > 0$.

13 INTEGRATION: AREAS AND VOLUMES

FORMULAE

Area: $\int y\,dx$; between curves $\int(\text{upper} - \text{lower})\,dx$. Volume about the x-axis $\pi\int y^2\,dx$; about the y-axis $\pi\int x^2\,dy$ (Jun 23, Jun 24).

$\int e^{kx}\,dx = e^{kx}/k$; $\int \sin kx\,dx = -\cos kx/k$; $\int \cos kx\,dx = \sin kx/k$; $\int 1/x\,dx = \ln x$.

- **Find intersections algebraically first**, then set the limits; for regions with a line, subtract a cone or triangle or integrate (line − curve).
- Square trig functions with double angles before integrating: $(3 + 2\sin x)^2$ needs $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ (Jun 22).
- Exponential volumes (Nov 20, Jun 21): $\pi\int e^{6x}\,dx = (\pi/6)e^{6x}$; give answers in π and e when asked.
- Exact areas with ln (Jan 23): keep ln 2 and surds; do not mix decimals with exact values.

14 FINAL A* CHECKLIST

1. Read the instruction: “use algebra”, “use calculus”, “using a formula from page 2”, “exact”, “hence”.
2. Inequalities: critical values, sketch, correct region, strict or non-strict, combined set.
3. Trig: never divide by sin or cos (factorise); all solutions in range, none outside; radians when asked.
4. Rational curves: asymptotes as equations, intercepts, both branches, dy/dx by the quotient rule.
5. Series: $|r| < 1$ before S_∞; least n with logs; reverse the inequality for $\log r < 0$.
6. Vectors: parallel sides as scalar multiples; collinearity by equating coefficients.
7. Calculus: justify max/min; substitute the right quantity in rates; reject negative times.
8. Integration: intersections first; integrated expression and limits visible; exact form.
9. Graphs: draw the straight line; read to one decimal place.
10. Leave 10 minutes for rechecking signs, brackets and the requested accuracy.